RESULTS — TMUA 2016 P1

14 / 20

20 attempted, 0 left blank · total time on paper 1:14:50

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TMUA SCORE

8.6 2024+ converted 8.6 Official

Official: TMUA 2016's published score conversion. 2024+ converted: our estimate of the equivalent on today's scale.

BY TOPIC

Area Integration	1 / 1
Circle Geometry	1 / 1
Differentiation	3 / 5
Exponentials and Logar…	2 / 2
General Number of Sol…	0 / 1
General Trigonometry	1 / 3
Geometry	1 / 2
Inequalities	0 / 1
Polynomial Expansions	1 / 2
Ratio and Proportion	2 / 2
Remainder Theorem	1 / 1
Sequences and Series	2 / 2
Transformation of Grap…	1 / 1
Trig Equation Number o…	1 / 1

BY DIFFICULTY

4.1	1 / 1
4.4	1 / 1
4.9	1 / 1
5.1	1 / 1
5.3	2 / 2
5.4	1 / 1
5.5	1 / 2
5.6	0 / 1
5.8	0 / 1
5.9	2 / 2
6.1	1 / 1
6.2	2 / 3
6.5	1 / 1
6.6	0 / 1
7.2	0 / 1

Per-question breakdown

#	YOUR ANSWER	CORRECT	RESULT	TIME		
1	A	H	✗	1:02	Bookmark	Review
2	E	E	✓	1:20	Bookmark	Review
3	C	C	✓	1:24	Bookmark	Review
4	B	B	✓	1:54	Bookmark	Review
5	C	C	✓	2:10	Bookmark	Review
6	C	C	✓	2:14	Bookmark	Review
7	B	B	✓	4:09	Bookmark	Review
8	B	F	✗	2:01	Bookmark	Review
9	D	D	✓	2:10	Bookmark	Review
10	E	E	✓	2:20	Bookmark	Review
11	E	E	✓	3:18	Bookmark	Review
12	E	E	✓	10:38	Bookmark	Review
13	A	C	✗	5:51	Bookmark	Review
14	D	D	✓	5:03	Bookmark	Review
15	C	C	✓	4:49	Bookmark	Review
16	C	C	✓	1:29	Bookmark	Review
17	B	D	✗	4:23	Bookmarked	Review
18	D	A	✗	2:23	Bookmark	Review
19	B	B	✓	6:30	Bookmark	Review
20	E	D	✗	9:42	Bookmark	Review

Questions I got wrong:

Calc mistake. didn't realise that there was a minus in front of px^2.

#	YOUR ANSWER	CORRECT	RESULT	TIME		
1	A	H	×	1:02	Bookmark	Hide

QUESTION

It is given that the expansion of $(ax+b)^3$ is $8x^3 - px^2 + 18x - 3\sqrt{3}$, where a, b and p are real constants.

What is the value of p?

A	$-12\sqrt{3}$ YOUR CHOICE	B	$-6\sqrt{3}$
C	$-4\sqrt{3}$	D	$-\sqrt{3}$
E	$\sqrt{3}$	F	$4\sqrt{3}$
G	$6\sqrt{3}$	H	$12\sqrt{3}$ CORRECT

SOLUTION

The correct solution is option H. For details see UAT official worked answers

Polynomial Expansions Difficulty 6.2

8 B F × 2:01 Bookmark Hide

QUESTION

Find the maximum angle x in the range $0^\circ \leq x \leq 360^\circ$ which satisfies the equation

$$\cos^2(2x) + \sqrt{3}\sin(2x) - \frac{7}{4} = 0$$

A	30°	B	60° YOUR CHOICE
C	120°	D	150°
E	210°	F	240° CORRECT
G	300°	H	330°

SOLUTION

The correct solution is option F. For details see UAT official worked answers

General Trigonometry Difficulty 5.6

$\sin 2x = A$

$1 - \sin^2(2x) + \sqrt{3}\sin 2x - \frac{7}{4} = 0$

$1 - A^2 + \sqrt{3}A - \frac{7}{4} = 0$

$-A^2 + \sqrt{3}A - \frac{3}{4} = 0$

$-4A^2 + 4\sqrt{3}A - 3 = 0$

$4A^2 - 4\sqrt{3}A + 3 = 0$ $\quad 4\sqrt{2} = \sqrt{32}$

$A = \frac{4\sqrt{3} \pm \sqrt{48-48}}{8}$

$= \frac{\sqrt{3}}{2}$

$\sin 2x = \frac{\sqrt{3}}{2}$

$0^\circ \leq 2x \leq 720^\circ$ ☆☆

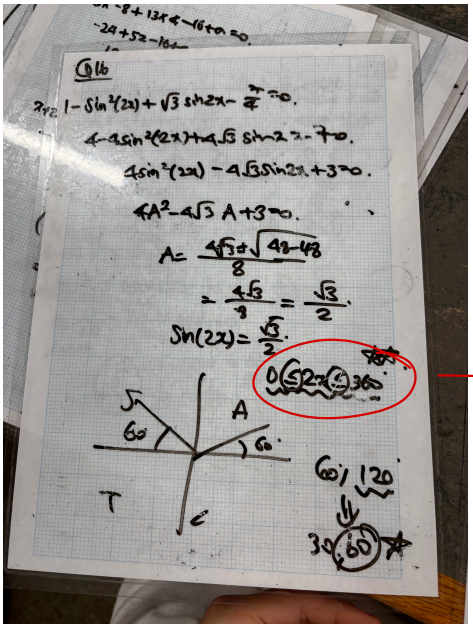
$\therefore \sin 60^\circ = \sin 120^\circ = \frac{\sqrt{3}}{2}$

$\sin(120 + 360) = \sin(480)$

~~$2x = 240$~~

$\therefore x = 240$ is max

Read the range WRONG



→ WRONG!!

should have done $0^\circ \leq 2x \leq 720^\circ$

need to learn how to do these Qs properly. Pops up every time.

13 A C × 5:51 Bookmark Hide

QUESTION

How many real roots does the equation $3x^5 - 10x^3 - 120x + 30 = 0$ have?

A 1 YOUR CHOICE B 2

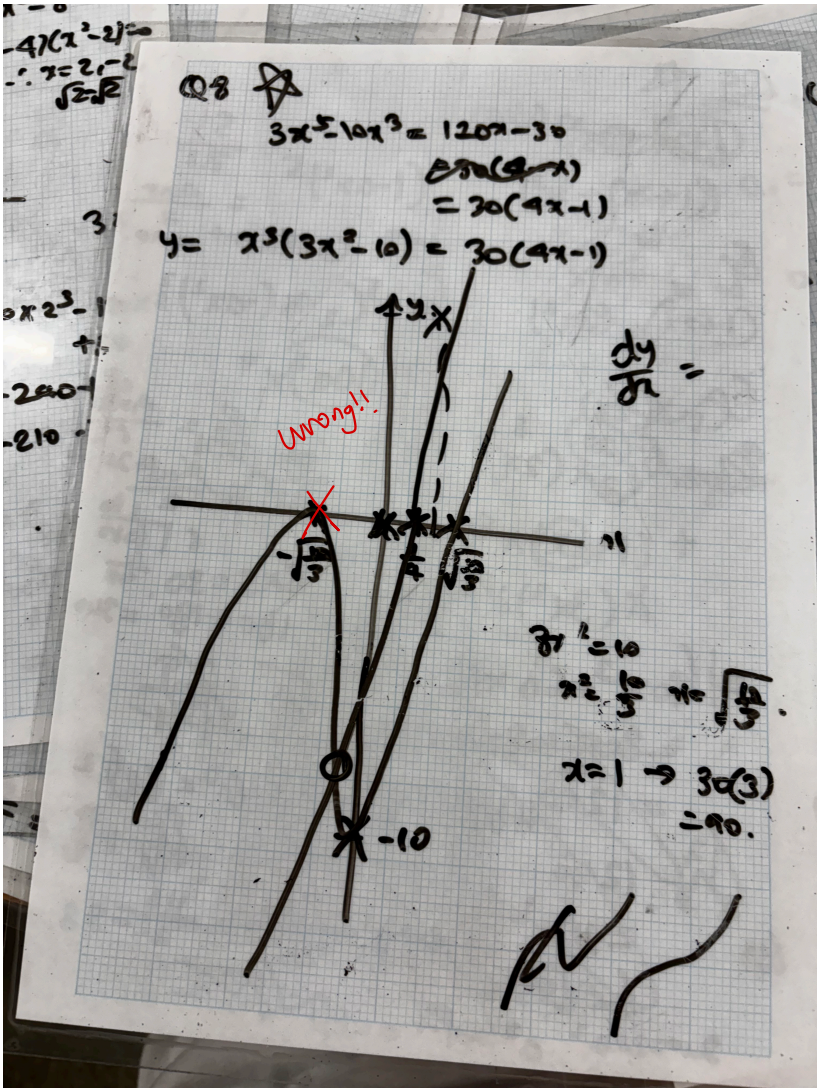
C 3 CORRECT D 4

E 5

SOLUTION

The correct solution is option C. For details see UAT official worked answers

Differentiation General Number of Solutions Difficulty 5.5



$\frac{dy}{dx} = 15x^4 - 80x^2 - 120$

$= 15(x^4 - 2x^2 - 8)$

$= 15(x^2 - 4)(x^2 + 2) = 0$

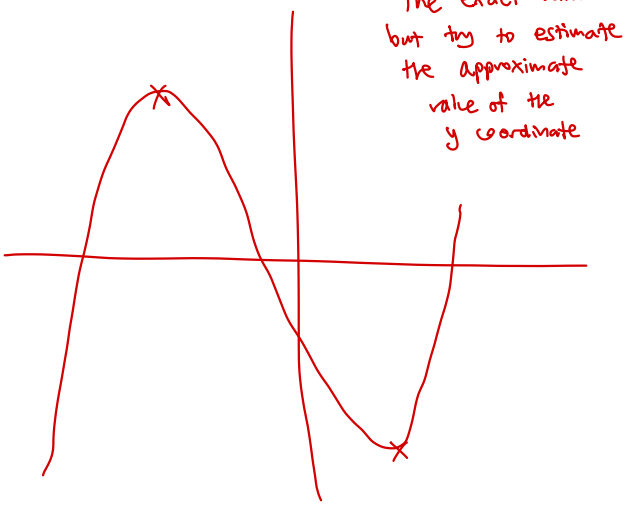
$\therefore x^2 - 4 = 0$

$\therefore x = 2$ or -2

$y = -36 + 80 + 240 + 30$ → $y > 0$

$y = 36 - 80 - 240 + 30$ → $y < 0$

DO NOT try to calculate the exact value but try to estimate the approximate value of the y coordinate



17 B D × 4:23 Bookmarked Hide

QUESTION

It is given that

$$y = (1 + 2\cos x)\cos 2x \quad \text{for } 0 < x < \pi$$

The complete set of values of x for which y is negative is

A $0 < x < \frac{\pi}{4}, \frac{2\pi}{3} < x < \frac{3\pi}{4}$

B $0 < x < \frac{\pi}{4}, \frac{3\pi}{4} < x < \pi$ YOUR CHOICE

C $0 < x < \frac{2\pi}{3}, \frac{3\pi}{4} < x < \pi$

D $\frac{\pi}{4} < x < \frac{2\pi}{3}, \frac{3\pi}{4} < x < \pi$ CORRECT

E $\frac{\pi}{4} < x < \frac{2\pi}{3}$

F $\frac{\pi}{4} < x < \frac{3\pi}{4}$

SOLUTION

The correct solution is option D. For details see UAT official worked answers

General Trigonometry Inequalities Difficulty 5.8

18 D A ✕ 2:23 Bookmark Hide

QUESTION

The function $\frac{1-x}{\sqrt[3]{x^2}}$ is defined for all $x \neq 0$.

The complete set of values of x for which the function is decreasing is

A $x \leq -2, x > 0$ CORRECT

B $-2 \leq x < 0$

C $x \leq 1, x \neq 0$

D $x \geq 1$ YOUR CHOICE

E $-2 \leq x \leq 1, x \neq 0$

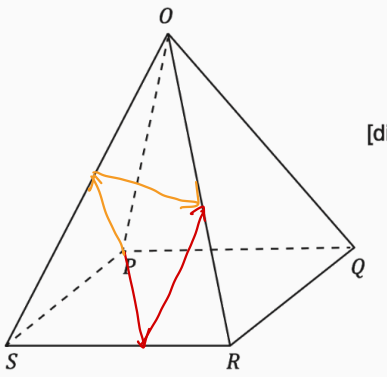
F $x \leq -2, x \geq 1$

SOLUTION

The correct solution is option A. For details see UAT official worked answers

20 E D × 9:42 Bookmark Hide

QUESTION



[diagram not to scale]

The diagram shows a square-based pyramid with base $PQRS$ and vertex O. All the edges of the pyramid are of length 20 metres.

Find the shortest distance, in metres, along the outer surface of the pyramid from P to the midpoint of OR.

A $10\sqrt{5-2\sqrt{3}}$

B $10\sqrt{3}$

C $10\sqrt{5}$

D $10\sqrt{7}$ CORRECT

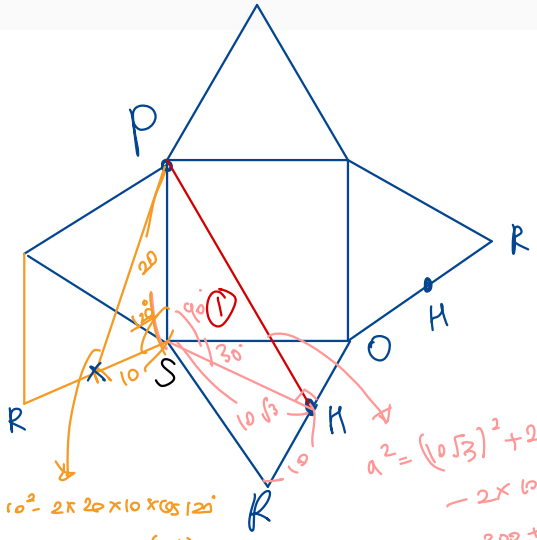
E $10\sqrt{5+2\sqrt{3}}$ YOUR CHOICE

SOLUTION

The correct solution is option D. For details see UAT official worked answers

Geometry Difficulty 7.2

think about different paths I could take to reach that point on the diagram!! ☆



$b^2 = 20^2 + 10^2 - 2\times 20\times 10\times\cos 120°$

$= 400 + 100 - 400\times\left(-\frac{1}{2}\right)$

$= 700$

$\therefore b = \sqrt{700} = 10\sqrt{7}$

$a^2 = (10\sqrt{3})^2 + 20^2 - 2\times 10\sqrt{3}\times 20\times\cos 120°$

$= 300 + 400 - 400\sqrt{3}\times\frac{1}{2}$

$= 700 - 200\sqrt{3}$